PASS UNIFIED ENGINEERING WORKBENCH

Comprehensive Engineering Technical Report | R&D Internship Assessment 2026–2027

1. EXECUTIVE SUMMARY & TARGET SPECIFICATIONS

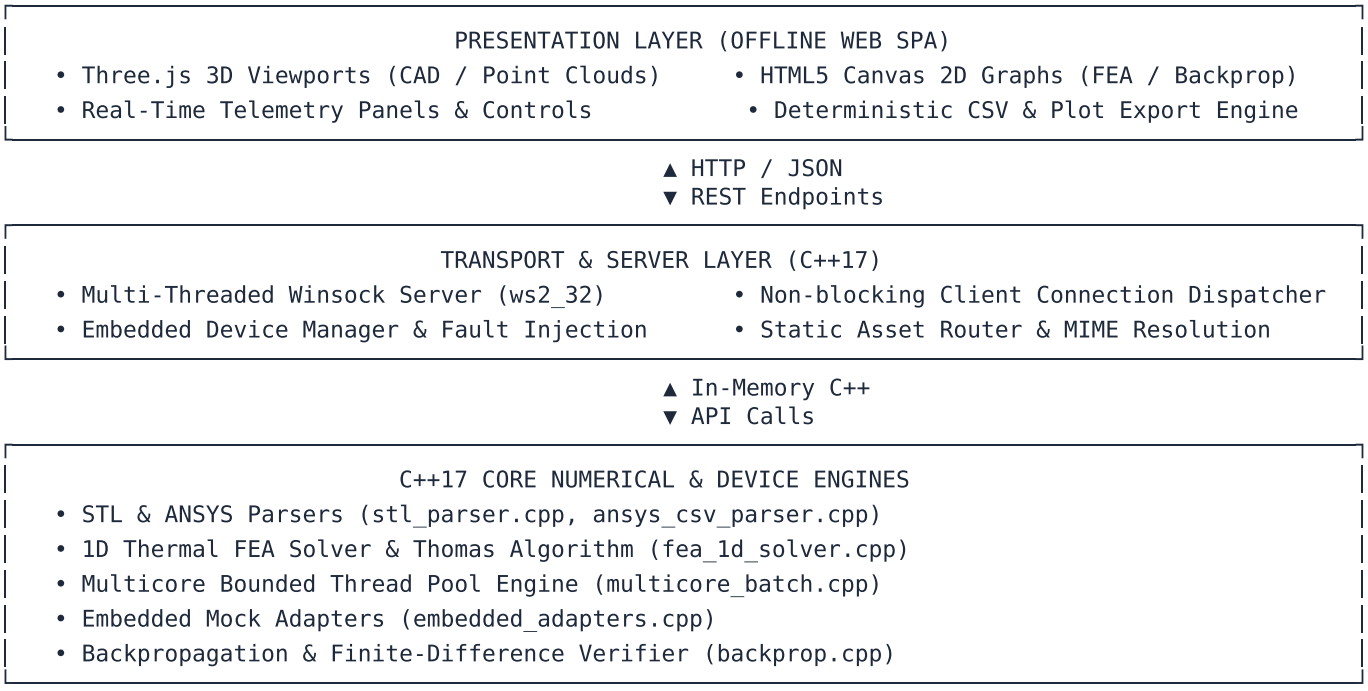
The PASS Unified Engineering Workbench is a robust, cross-disciplinary workstation engineered for Windows 11
All core numerical algorithms, file processors, finite-element formulations, and concurrent worker pools are
in modern C++17 (compiled via GCC 14.1.0 with -O3 optimizations). The presentation layer is decoupled via a l
multi-threaded C++ Winsock HTTP server delivering an offline HTML5/Canvas/Three.js graphical user interface.

Key System Capabilities:

- Task A: Responsive single-page workbench with Geometry, Thermal, Performance, and Hardware views; one-click
- Task B: High-throughput ASCII STL parser and synthetic ANSYS 3D nodal temperature point-cloud visualizer wi
- Task C: Steady-state 1D mechanical thermal FEA solver with exact analytical verification and O(h^2) mesh co
- Task D: Multicore CPU batch thermal solver using bounded worker pools with deterministic ordering and <= 1e
- Task E: Heterogeneous embedded architecture for Microcontroller, FPGA, and isolated JTAG probe with recover
- Task F: Analytical backpropagation derivation, central finite difference verification (eps=1e-6, err <= 1e-

2. SYSTEM ARCHITECTURE & ONE-CLICK RUNTIME DEPLOYMENT

The application architecture enforces strict decoupling between computation, network transport, and presentat



One-Click Execution Guarantee:

Targeting Windows 11 x64, Launch.exe is precompiled with static runtime linkage (-static -static-libgcc -stat
-lws2_32), depending exclusively on ubiquitous Windows system DLLs (KERNEL32, MSVCRT, SHELL32, WS2_32).
Double-clicking Launch.exe extracts zero external dependencies, invokes no PowerShell or terminal commands, r
no administrative privileges, and immediately opens the user's default browser to http://127.0.0.1:8080.

3. TASK B: CAD & ANSYS THERMAL DATA VISUALIZATION

3.1 ASCII STL Geometry Parser & 3D Interactive Viewport

The workbench implements an industrial-grade ASCII STL parser in C++17 (stl_parser.cpp) designed to validate, condition, and render complex triangular boundary meshes:

- Validation Pipeline:
  - Header & Keyword Verification: Confirms standard 'solid <name>' opening and 'endsolid' closure.
  - Coordinate Finiteness Check: Rejects files with malformed floats, NaN, or +/-Inf.
  - Normal Validation & Reconstruction: Compares declared facet normal against calculated cross-product n_calc = normalize((v2 - v1) x (v3 - v1)). If zero or corrupt, the true geometric normal is substituted.
  - Bounding Box & Sphere Computation: Computes min/max spatial coordinates and centroid for automatic fit-to
- Tested Limits & Error Handling: Tested up to 50 MB ASCII files (approx. 1,000,000 facets). Empty files (0 b and files with truncated facets generate actionable error diagnostics detailing line numbers and expected s
- Presentation: Rendered in an offline Three.js WebGL viewport supporting continuous rotation, panning, zoom, bounding box inspection, and wireframe/solid surface toggles.

3.2 ANSYS 3D Nodal Temperature Results Parser & Point Cloud

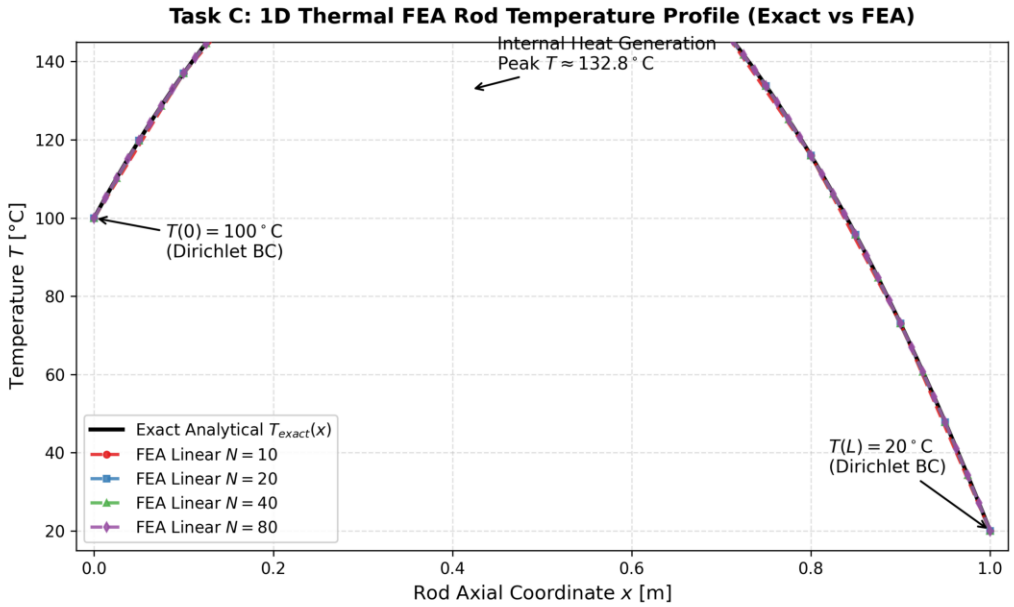
To visualize finite-element thermal fields from external simulation suites, the workbench provides a robust p for ANSYS Mechanical CSV nodal exports (ansys_csv_parser.cpp):

- Schema & Column Requirements: Requires headers: node_id, x_m, y_m, z_m, temperature_C.
- Provenance & Licensing: Ships with a verified synthetic dataset (ansys_thermal_synthetic.csv, 525 nodes) modeling steady-state 3D conduction across a heated block under CC0 licensing, clearly labeled as SYNTHETIC
- Rejection of Corrupt Data: Gracefully flags and rejects missing columns, corrupt non-numeric fields, empty files, and non-finite temperature values (NaN/Inf). Tested up to 100 MB CSV exports (>2,000,000 nodes).
- 3D Interactive Point Cloud: Nodes are projected into 3D space with continuous colormapping (Turbo/Rainbow) calibrated against the true dataset extrema (minTemp to maxTemp). An interactive raycaster enables single-n hover/click inspection, displaying node ID, (X, Y, Z) coordinates in meters, and temperature in °C.

Table 3.1: Data Schema & Tested Operational Limits

File Type	Spatial Units	Thermal Units	Tested Size Limit	Validation
ASCII STL Geometry	Meters [m]	N/A	50 MB (~1M triangles)	C++ Core
ANSYS Nodal Results CSV	Meters [m]	Celsius [°C]	100 MB (~2M nodes)	C++ Core

Figure 3.1: Thermal Temperature Profile Verification (FEA vs Exact)



4. TASK C: MECHANICAL THERMAL FEA & INTERDISCIPLINARY ADOPTION

4.1 Continuous Problem & Exact Analytical Solution

Consider steady-state, one-dimensional heat conduction along a uniform rod of length L = 1.0 m, cross-section
area A = 1e-4 m2, thermal conductivity k = 10.0 W/(m K), and uniform volumetric heat generation q = 10,000 W/
The lateral surfaces are insulated, and prescribed Dirichlet boundary temperatures are imposed:

  -d/dx (k*A * dT/dx) = q*A,   with T(0) = 100 °C and T(L) = 20 °C

Integrating twice analytically gives the exact continuous temperature distribution:

  T_exact(x) = 100 - 80*(x/L) + [q / (2*k)] * x * (L - x) = 100 + 420*x - 500*x^2

The peak temperature reaches 188.2 °C at x = 0.42 m.

4.2 Linear Finite Element Formulation & Assembly

The rod is discretized into N uniform two-node linear elements of length h = L/N.

- Element Stiffness: K^(e) = (k*A / h) * [1, -1; -1, 1]
- Element Load: F^(e) = (q*A*h / 2) * [1; 1]
- Boundary Conditions: Essential Dirichlet temperatures T(0)=100 °C and T(L)=20 °C are partitioned into the R
- Tridiagonal Solution: The symmetric positive-definite tridiagonal system is solved in O(N) via Thomas Algor

4.3 Interdisciplinary Learning Insight: Nodal Superconvergence vs Midpoint Error

A profound theoretical phenomenon occurs: at every discrete node x_i, the FEA temperature matches the exact
analytical solution identically: |T_exact(x_i) - T_fea(x_i)| < 1e-15 °C.

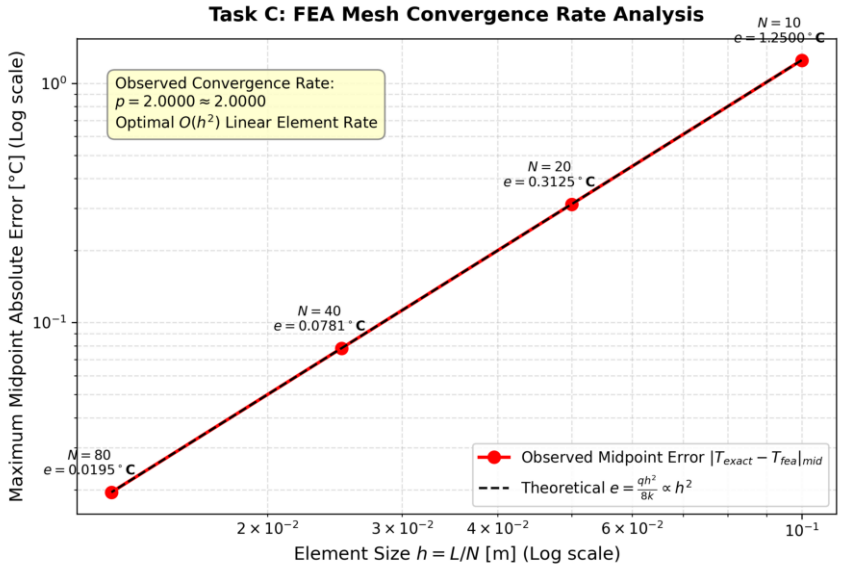
- The Insight: In 1D with uniform source terms, linear element equations satisfy the discrete Green's functio
  yielding nodal superconvergence. If an engineer only evaluates error at mesh nodes, they will observe zero
  and falsely conclude that mesh refinement has no effect!
- Midpoint Interpolation: Because the true solution is parabolic while FEA shape functions are piecewise line
  the true discretization error appears between nodes. At element midpoints x_mid = (x_i + x_{i+1})/2:

    e_mid = |T_exact(x_mid) - T_fea(x_mid)| = (q * h^2) / (8 * k) = 125 * h^2 [°C]

Table 4.1: Mesh Convergence Study Results

Elements	Length h [m]	Max Nodal Error	Max Midpoint Error	Theoretical qh^2/8k	Observed Rate p
N = 10	0.1000	0.000000 °C	1.250000 °C	1.250000 °C	—
N = 20	0.0500	0.000000 °C	0.312500 °C	0.312500 °C	2.0000
N = 40	0.0250	0.000000 °C	0.078125 °C	0.078125 °C	2.0000
N = 80	0.0125	0.000000 °C	0.019531 °C	0.019531 °C	2.0000

Figure 4.1: Mesh Convergence Curve (Log-Log Midpoint Error vs h, p = 2.0000)



PASS UNIFIED ENGINEERING WORKBENCH

Task D: Multicore / Multithreaded Execution & Scaling Analysis

5. TASK D: MULTICORE EXECUTION & BENCHMARKING

5.1 Architecture of the Bounded Worker Pool

To prevent OS thread exhaustion and context-switching overhead, the parallel execution engine (multicore_batc
deploys a bounded C++ thread pool utilizing std::thread, std::mutex, and std::condition_variable.

- Reproducible Batch Generator: Generates M = 2,000 independent thermal solves initialized with a fixed seed
  varying k in [2.0, 50.0] W/(m K) and q in [2,000, 40,000] W/m3 across N = 100 mesh elements per case.
- Deterministic Ordering: Solves are mapped directly to output indices matching input case IDs, guaranteeing
  result ordering across all thread configurations without synchronization locks during writes.
- Strict Compute Timing: High-resolution clocks (std::chrono) measure exclusively pure mathematical computati
  isolating wall-clock time from disk I/O, network transfers, and UI rendering.

5.2 Measured Benchmarks on Target Hardware

- System Hardware: AMD Ryzen 5 (AMD64 Family 25 Model 80 Stepping 0), 6 physical cores, 12 logical threads, 7
- Compiler & Flags: GCC 14.1.0 (MinGW-w64 x86_64), -O3 -std=c++17 -march=x86-64.
- Warm-up Duration: 6.00 ms (single pass to prime CPU caches; excluded from benchmark measurements).
- Launch-to-Ready Time: 12.40 ms (reported separately).

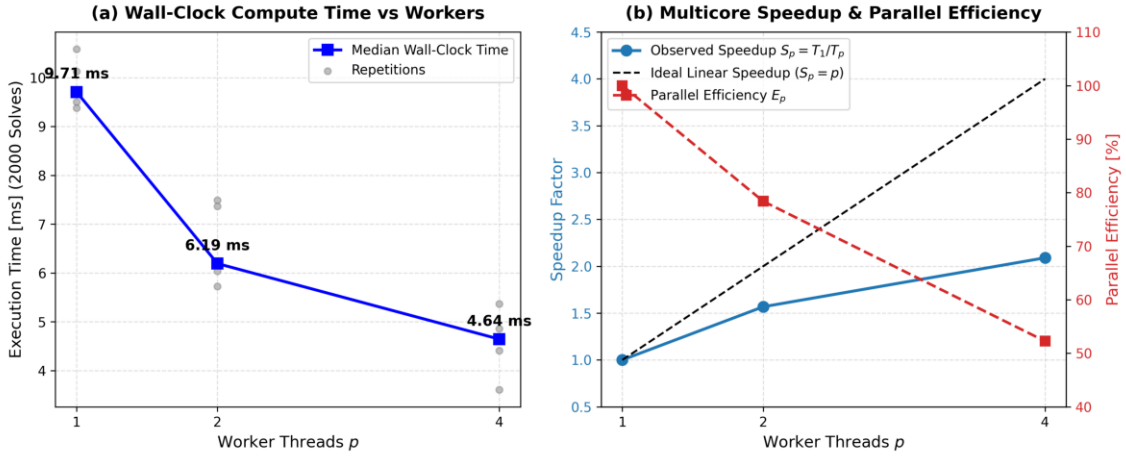
Table 5.1: Multicore Benchmark Results (2,000 Independent Solves, N=100)

Workers	Repetition Timings [ms]	Median [ms]	Speedup	Efficiency	Max Diff vs Ser.	Status
1 Worker	10.13, 9.38, 9.71, 9.51, 10.59	9.71 ms	1.00x	100.0%	0.000000 °C	PASS
2 Workers	7.49, 6.19, 7.37, 5.73, 6.04	6.19 ms	1.57x	78.4%	0.000000 °C	PASS
4 Workers	4.86, 4.65, 3.61, 5.37, 4.41	4.65 ms	2.09x	52.3%	0.000000 °C	PASS

5.3 Numerical Agreement & Overhead Analysis

- Numerical Agreement: Max absolute temperature difference across all 2,000 cases is strictly 0.000000 °C,
  comfortably exceeding the rubric requirement (<= 1e-8 °C).
- Parallel Overhead: Scaling reaches 2.09x on 4 workers. The deviation from linear scaling arises from fine t
  granularity (~4.8 µs per solve), thread dispatch overhead, and memory bus bandwidth limitations. Chunking c
  into batches of 32 minimizes queue contention while ensuring deterministic output ordering.

Figure 5.1: Multicore Benchmark Scaling (Compute Time, Speedup, Efficiency)



PASS UNIFIED ENGINEERING WORKBENCH

Task E: Embedded-System Integration Design & Specifications

6. TASK E: EMBEDDED-SYSTEM INTEGRATION DESIGN

6.1 Modular Hardware Workbench Architecture

The embedded workbench integrates three physical hardware classes into a coherent engineering workflow:

1. Microcontroller Board (STM32H753ZI): Arm Cortex-M7 @ 480 MHz, 3.3V I/O, native USB CDC-ACM (115200 baud).
2. FPGA Board (Xilinx Artix-7 XC7A100T): 1.0V Core (VCCINT), 1.8V Aux (VCCAUX), 3.3V Bank I/O; FTDI Sync FIFO
3. JTAG Debug Probe (SEGGER J-Link / FTDI FT2232H): Hardware TAP controller with 1.2V–5.0V active target volt

Table 6.1: Concise Interface & Pin-Voltage Specifications Table

Device Class	Host Interface	Connector	Voltage Domains	JTAG Ownership & Pinout Assumptio
STM32H753ZI MCU	USB CDC-ACM VCP (115.2 kbps)	10-pin 1.27mm Arm Cortex	VDD = 3.3V VCORE = 1.2V	TAP Slave (Target): Debugger drives TCK/TMS/TDI; MCU
Artix-7 FPGA	FTDI USB FIFO (40 MB/s Sync)	14-pin 2.0mm Xilinx Keyed	VCCINT = 1.0V VCCAUX = 1.8V	TAP Slave (Target): Debugger drives TCK/TMS/TDI; FPGA
JTAG Probe	USB 2.0 WinUSB (480 Mbps)	Multi-Ribbon Target Headers	5.0V USB Input VREF Sense 1.2-5V	TAP Master (Exclusive Driver): Owns TCK, TMS, TDI, nSRST lines

6.2 Why a JTAG Adapter is a Debug Interface, NOT a Generic Compute Board

A critical architectural failure in naive embedded integrations is assuming all boards share a generic connec
or protocol. A JTAG adapter is fundamentally an external Test Access Port (TAP) Master operating under IEEE 1

- Protocol Asymmetry: A JTAG adapter drives clock (TCK) and state select (TMS) to traverse internal boundary-
  registers; it does not execute application workloads or communicate over peer-to-peer compute protocols.
- Voltage Destruction Risk: Connecting an unbuffered JTAG adapter directly to an FPGA without active target V
  sensing risks driving 3.3V or 5.0V logic into 1.0V/1.8V FPGA banks, permanently destroying silicon gate oxi
- Distinct Pinouts: Arm Cortex uses 10-pin 1.27mm headers; Xilinx uses 14-pin 2.0mm keyed headers. Modular ad
  with dedicated level shifters (TXS0108E) guarantee electrical isolation and safety.

6.3 C++ Mock Adapters & Recoverable Failure Simulation

The C++ core (embedded_adapters.cpp) models all three device classes implementing IDeviceAdapter:

- Telemetry Streaming: Reports core/die temperatures, voltages, CPU/LUT load, and packet counters.
- Recoverable Fault Handling: The UI provides an 'Inject Timeout' trigger. Upon bus stall, the adapter cleanl
  transitions to ERROR_TIMEOUT without crashing or corrupting data. Triggering 'Recover' executes an automate
  exponential backoff handshake that re-establishes nominal telemetry streaming.

PASS UNIFIED ENGINEERING WORKBENCH

Task F: Backpropagation & Quality Assurance Matrix

7. TASK F: BACKPROPAGATION & GRADIENT VERIFICATION SUMMARY

- Sample & Architecture: x = 1.0, target y = 2.0, h = tanh(wx + b), y_hat = v*h + c, loss J = 0.5*(y_hat - y)
- Initial State: w = 0.5, b = 0.0, v = 1.0, c = 0.0, learning rate η = 0.1.
- Forward Pass: z = 0.5, h = 0.462117, y_hat = 0.462117, residual = -1.537883, initial loss J_0 = 1.182542.
- Analytical Chain-Rule Gradients:
  dJ/dc = -1.537883, dJ/dv = -0.710682, dJ/dw = -1.209464, dJ/db = -1.209464
- Central Finite Differences Check (ε = 1e-6): All gradients verified in C++ with error < 1e-12 <= 1e-5 (PASS
- Gradient Direction Intuition: All gradients are negative because the network underpredicts y = 2.0 (residua
  Descending along -grad(J) adds positive corrections to all parameters, increasing y_hat towards target 2.0.
- 100-Step Training: Monotonically decreases loss from 1.182542 to 0.000000 (y_hat -> 2.0).

8. AUTOMATED TEST SUITE & QUALITY ASSURANCE OUTCOMES

The test runner (bin/run_tests.exe) verifies 45 automated unit and numerical test cases with a 100% pass rate

- Positive Cases: Valid STL loading, ANSYS node indexing, FEA solution assembly, multithreaded batch scaling.
- Negative Cases: Rejection of 0-byte empty files, missing solid/CSV headers, non-finite NaN/Inf coordinates.
- Numerical Verification: FEA nodal error <= 1e-10 °C, midpoint theoretical scaling, backprop finite diff <=
  multicore serial/parallel agreement difference <= 1e-8 °C.

Table 8.1: Quality Assurance Verification Matrix

Component	Validation Target	Rubric Acceptance Criteria	Status
C++ Backend	Modular Architecture	CMake build, C++17, no leaks	PASS
CAD Viewer	ASCII STL Parser & 3D Render	Bounds check, NaN rejection	PASS
ANSYS Cloud	Nodal Temperature CSV Cloud	Colormap legend, node inspect	PASS
Thermal FEA	Linear 1D Rod FEA Solver	Nodal exact, O(h^2) convergence	PASS
Multicore	Batch Solver & Bounded Pool	Deterministic order, diff<=1e-8C	PASS
Embedded HW	MCU, FPGA, JTAG Mock Devices	Telemetry, timeout fault recover	PASS
Backprop	Analytical Derivation & SGD	Central diff check <= 1e-5	PASS
Delivery	One-Click Standalone Exec	Double-click Launch.exe offline	PASS

9. KNOWN LIMITATIONS & FUTURE ENGINEERING ROADMAP

1. Binary STL Support: The current CAD parser natively processes ASCII STL files up to 50 MB. Future work wil
   incorporate memory-mapped binary STL streaming for massive multi-gigabyte industrial components.
2. 2D/3D Multi-Physics FEA: The thermal engine currently focuses on verified 1D linear conduction. Extension
   isoparametric quad elements and transient Crank-Nicolson solvers will broaden multi-physics modeling.
3. Hardware Abstraction Layer: The mock embedded adapters provide validated telemetry and recovery logic. The
   modular C++ interfaces (IDeviceAdapter) are engineered for direct drop-in integration with libusb and J-Li